

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **software validation methodologies, Equipment & Software Qualification processes, and compliance documentation** across enterprise Java/Spring Boot applications and production systems at HCL, ensuring adherence to **Global & Site Policies, SOPs, and regulatory requirements** with **100% QA audit compliance**.
- Established **Non-Device Software Lifecycle processes** for Equipment with Software, Software Only Systems, Data Reporting Software, and Spreadsheet Validation across multiple project teams; reviewed and approved all **Equipment & Software Qualification deliverables (QP, PRA/EHA, EFS, EQP, SCS, EQR)** to ensure systems conformed to **regulatory, quality system, and technical requirements**.
- Managed **Change Control processes** for Equipment & Software qualification projects using **Jira, Git, and Jenkins CI/CD pipelines**; ensured all new Equipment & Software updates and system changes were qualified and documented according to established **BSC Change Control procedures**, reducing qualification cycle time by **18%**.
- Identified, prioritized, and resolved **discrepancies and deviations** in Equipment & Software Qualification projects through **systematic problem-solving methodologies** and comprehensive risk assessment; maintained **defect tracking** documentation in **Jira** with full traceability and regulatory evidence, achieving **95% first-pass validation success rate**.
- Delivered **software validation/compliance/Part 11/data integrity guidance** aligned with **FDA CFR Part 11, 21 CFR Part 11, IEC 62304, and ISO 13485** standards; trained cross-functional teams across production, engineering, and operations on **Equipment & Software validation best practices** and **Right First Time** culture, improving team validation competency by **40%**.
- Drove **automation in Software Quality department processes** using **Python scripting, SQL data validation, and REST API integrations** across **PostgreSQL, MySQL, and Oracle** databases; automated validation report generation and evidence collection, reducing manual QA overhead by **22%** and accelerating time-to-compliance for Product Release Authorization.

SKILLS

Software Quality & Validation – Equipment & Software Qualification, Non-Device Software Lifecycle Processes, Software Validation Methodologies, Change Control Management, Discrepancy & Deviation Management, Right First Time Culture, Best Practices Documentation.

Regulatory & Compliance – FDA CFR Part 11, 21 CFR Part 11, IEC 62304, ISO 13485, Global & Site Policies, SOPs, WIs, Product Release Authorization (PRA), Equipment Qualification Plan (QP), Validation Reports (EQR).

Software Development & Testing – Java Spring Boot, REST APIs, Python Scripting, SQL (PostgreSQL, MySQL, Oracle), Data Validation, System Integration Testing, UAT Coordination, Automated Test Reporting.

Tools & Platforms – Jira, Confluence, Git, Jenkins CI/CD Pipelines, CloudWatch, Splunk, Dynatrace, Docker, AWS (EC2, S3, Lambda), Linux, Data Quality Automation.

Technical Documentation & Training – Quality Plans, Test Protocols, Validation Reports, Risk Assessment Documentation, Standard Operating Procedures, User Guides, Cross-Functional Team Training, Competency Development Programs.

Business & Quality Improvement – Process Automation, Problem-Solving Methodologies, Defect Tracking & Traceability, Project Scheduling, Vendor/System Collaboration, Cost-Effectiveness Optimization, Patient & Product Safety Focus.

Data & Reporting – Data Integrity Assurance, Compliance Reporting, Spreadsheet Validation, Evidence Collection, Audit Trail Management, Performance Metrics & KPI Tracking.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Medical Device Software Validation & Compliance Platform (HCL Technologies)

- Architected and deployed a centralized **Equipment & Software Qualification management system** supporting **Non-Device Software Lifecycle processes** including Software Only Systems, Data Reporting Software, Spreadsheet Validation, and AI Applications across 8+ product development and operational teams.
- Automated **validation deliverable generation, compliance documentation, and Change Control workflow** using **Python ETL pipelines, REST API integrations, and SQL data validation** across **PostgreSQL and Oracle databases**; integrated with **Jira and Git workflows** to enforce **FDA CFR Part 11, IEC 62304, and ISO 13485 compliance requirements**.
- Implemented **automated risk assessment and deviation tracking system** with real-time audit trails, reducing manual discrepancy documentation effort by **35%** and ensuring **100% regulatory traceability** for Product Release Authorization (PRA/EHA/EFS/EQP/SCS/EQR deliverables).
- Reduced Equipment & Software Qualification cycle time from **8 weeks to 6.5 weeks** by automating evidence collection, compliance reporting, and sign-off workflows; delivered **45+ concurrent validation projects** with zero compliance violations.
- Trained and coached **12+ cross-functional engineers** on **software validation methodologies, Part 11 data integrity requirements, and Right First Time practices**; established **Quality Engineering Center of Excellence** documentation and playbooks, improving first-pass validation success rate to **92%**.

SWIFT Payment Operations Validation & Compliance Dashboard (HCL Technologies)

- Developed **Equipment & Software Qualification and compliance framework** for a mission-critical **SWIFT messaging and payments-operations dashboard** supporting ISO 20022 message flows, SEPA instant payments, and cross-border payment processing with **SLA-driven production monitoring** using **Splunk and CloudWatch**.
- Established **software validation protocols and regulatory compliance documentation** aligned with **PSD2, GDPR, and financial-services audit trail requirements**; ensured all system updates and configuration changes followed **established Change Control procedures** with full Quality Assurance sign-off.
- Automated **validation evidence collection and compliance reporting** via **Python scripts and SQL queries** on **PostgreSQL** transaction databases, reducing manual audit preparation time by **28%** while maintaining **100% regulatory compliance** for financial-services audits and regulatory inspections.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)

- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)